

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd May 2012

Subject CT4 – Models

Time allowed: Three Hours (10.00 – 13.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
4. *In addition to this paper you will be provided with graph paper, if required.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** **a)** You have been asked to advise on a study of human sickness with three states – healthy, sick and dead. Comment on the ease of extending the following models to multiple states described in the study:
- i)** Markov multiple state model
 - ii)** Poisson model
 - iii)** Binomial model
- (3)
- b)** List two other criteria that you might use to compare the above models.
- (2)
- [5]**
- Q. 2)** **a)** State, in words, the consistency condition between the random variable representing future lifetimes from different ages, and provide a mathematical representation for the same.
- For certain types of lung cancers, the prognosis is generally poor. Following complete surgical resection, the five-year survival probability is only about 14%. However, the first 18 months are considered critical and if the patient survives this initial time, the survival probability for the remainder of five years goes up to 77%.
- (3)
- b)** Use the consistency condition above to determine the survival probability of patients for the first 18 months following a surgery.
- (1)
- c)** A fellow student has commented that given the first 18 months are the most critical, the probability of survival for these 18 months should be less than 14%. Given your answer above, or otherwise, state whether the student's assertion is correct or not. Clearly explain the rationale behind your answer.
- (2)
- [6]**
- Q. 3)** You are a student Actuary and recently joined the pricing team of a life insurance company. You have been asked to develop a pricing model for a term assurance product.
- a)** Describe the key steps for construction and use of a model for this purpose.
- (6)
- b)** State with reasons why you would want to perform sensitivity testing.
- (2)
- Note: A term assurance product is a life insurance product that provides for an insurance cover for a fixed term. If the life insured dies during the policy term, the insurer pays out a fixed sum assured to the nominee of the deceased. No benefit is payable upon the expiry of the policy term if the life insured survives the policy term.*
- [8]**

- Q. 4)** Clients arrive at a tax consultant's office according to a Poisson process with the rate $\lambda = \frac{1}{10}$ per minute.

The consultant will not meet a client until at least three clients are in the waiting room. In other words, at any given point of time, there should be at least two clients waiting in the waiting room when the consultant proceeds to meet a client. After the consultant has met 15 clients on a given day, he is happy and meets all the arriving clients irrespective of the number of clients waiting.

- a) Calculate the expected waiting time until the consultant meets the first client. (2)
- b) What is the probability that consultant does not meet any client in the first two hours? (2)
- c) Today, the consultant wants to attend a family function after three hours. However, he wants to meet at least 8 clients today before he proceeds to attend the function. What is the probability that he meets at least 8 clients in three hours? (3)

[7]

- Q. 5)** Define rate interval and clearly specify the exact ages to which the rate interval applies if the age labels is defined as:

- age nearest birthday
 - age next birthday
- (2)

A bulk annuity provider carries out valuation each 31 December and records in force policies using age label "age next birthday". You are provided with the number of policies in-force for the last five valuations between ages 71-75:

Number of policies in-force as at the valuation date

	<i>Age next birthday</i>				
	71	72	73	74	75
31-Dec-07	45,780	44,921	43,680	42,890	41,920
31-Dec-08	46,352	44,819	43,955	42,719	41,904
31-Dec-09	46,485	45,379	43,855	42,988	41,737
31-Dec-10	47,293	45,509	44,403	42,890	41,999
31-Dec-11	48,749	46,295	44,530	43,426	41,904

New annuities vest at exact age 70 only. Assuming that death is the only cause of exit for the given annuitants, calculate estimates for initial mortality rate (q) for lives aged 71, 72 and 73 last birthday.

(6)

[8]

- Q. 6)** Steve Jobs' biography is very popular these days at a college library. Keeping in mind the demand, the librarian wants to understand students' reading patterns and in particular how these influence the availability of this book at the library.

The librarian had read a book on Markov Jump Processes and realized these can be applied for his analysis. He decides to use a two-state Markov Process with the states being reflecting the status of the book - borrowed (B) or available (A).

The fortnightly transition rates between these states are given below:

$$\begin{array}{cc} & \begin{array}{cc} A & B \end{array} \\ \begin{bmatrix} -0.9 & 0.9 \\ 0.1 & -0.1 \end{bmatrix} \end{array}$$

The probabilities of the book being either borrowed or available at time t are denoted by $P_B(t)$ and $P_A(t)$.

- a) Write down Kolmogorov's forward equation for $P_B'(t)$. (2)
- b) Solve the equation in part (a) to obtain a formula for the probability that the book is borrowed at time t , given that it was available at time 0. (3)
- c) By using an indicator function or otherwise, calculate the expected period of time the book is available over the period $[0, t]$ and hence calculate the expected number of days the book is available over a fortnight (15 days). (4)

[9]

- Q. 7)** A PC hardware manufacturer (XYZ Ltd) is assessing the potential behaviour of corporate buyers in allocating their budgets to various companies. High level investigations have shown that corporate buyers usually have several companies on their vendor list, but that they prefer to follow a target percentage to be bought from each vendor. Each year, as new innovations and pricing are announced, these buyers reassess their target percentage for each vendor.

An internal research indicates that, for the purpose of high level analysis, a corporate buyer's target allocation for XYZ Ltd can be one of 100%, 75%, 65% or 25% of its budget.

A detailed study of previous buying behavior has produced the transition matrix as given below for the probability of changing each year between target percentages:

$$\begin{array}{cccc} \mathbf{100\%} & \mathbf{75\%} & \mathbf{65\%} & \mathbf{25\%} \\ \begin{bmatrix} 0.6 & 0.3 & 0.1 & 0 \\ 0 & 0.7 & 0.3 & 0 \\ 0 & 0.4 & 0.4 & 0.2 \\ 0 & 0.2 & 0.5 & 0.3 \end{bmatrix} \end{array}$$

Presently, out of every 100 corporate buyers, 5 have a target percentage for XYZ Ltd of 100%, 30 a target percentage of 75%, 45 a target percentage of 65% and 20 a target percentage of 25%.

- a) Draw the state transition diagram. (2)
- b) What will be the percentage of corporate buyers having a target percentage for XYZ Ltd of 65%:
- i) in 2 years time; and
- ii) over the long-run (6)
- c) Would you expect the actual percentage of corporate buyers having a target percentage for XYZ Ltd of 65% to approach the long-run figure calculated above and why? (2)

[10]

- Q. 8)** Two friends, Lalit and Amit, are playing a game of coin by exchanging marbles. Presently, Lalit has 1 marble and Amit has 2 marbles.

The game involves tossing a fair coin every hour. If the outcome is heads, Lalit gives a marble to Amit; and if the outcome is tails Amit gives a marble to Lalit. The game is continued till either of the players collects all three marbles. Let X_n denote the number of marbles held by Lalit after n hours.

- a) Define a Markov Chain and show that X_n is a Markov chain. (3)
- b) Show that for even n (i.e. $n = 2k$ where k is an integer) (5)

$$p_{11}(n) = \left\{\frac{1}{2}\right\}^n = p_{22}(n)$$

$$p_{10}(n) = \frac{2}{3} \times \left[1 - \left(\frac{1}{4}\right)^k\right] = p_{23}(n)$$

$$\text{where } p_{ij}(n) = P(X_n = j \mid X_0 = i)$$

- c) Determine the n -step transition probability matrix for even n using above. (6)

[14]

- Q. 9)** Officials from an international lion-breeding program are considering crossbreeding Asiatic and African lions, with the aim improving the genetic pool and increasing the average life span of lions. They have provided you with the following notes having observed a group of lions over a period of two years:

Name of Lion	Sub-specie	Notes
Simba	African	Alive, aged 9 years at the end of observation period
Mufasa	African	Died at the age of 9 years
Nala	African	Alive, aged 8 years at the end of observation period
Bhasuraka	Asian	Died at the age of 10 years
Sarabi	African	Alive, aged 13 years at the end of observation period
Scar	African	Died at the age of 12 years
Ajradanstra	Asian	Alive, aged 8 years at the end of observation period
Karalakesara	Asian	Alive, aged 11 years at the end of observation period

The following model has been suggested for the force of mortality:

$$\mu(x | Z = z) = \mu_0(x)e^{\beta z}, \text{ where:}$$

- x denotes the age of lion
- $\mu_0(x)$ is the base line hazard
- $z=0$ for African lions, $z=1$ for Asian lions.

a) Calculate the maximum partial likelihood estimate of the model parameter, β and thus determine the relative hazard rates of Asian and African lions. (6)

b) Construct an approximate 95% confidence interval for the model parameter, β and thus comment on the significance of using different lion sub-species in the breeding program to improve longevity. (4)

c) The breeding program officials have noted that the life spans of lions are considerably improved if bred in captivity as opposed to in the wild. Therefore, they have asked you to expand your model to incorporate this in your model such that:

$$\mu(x | Z = \underline{z}) = \mu_0(x)e^{\beta^T \underline{z}}, \text{ where:}$$

- x denotes the age of lion
- $\mu_0(x)$ is the base line hazard
- $\underline{z}$ is a vector of covariates, where
 $z_1=0$ if African, 1 if Asian
 $z_2=0$ if in the wild, 1 if bred in captivity
 $z_3 = z_1 \times z_2$
- β is a vector of parameters, where
 β_1 is same as estimate calculated in part (i)
 $\beta_2 = -0.55$
 $\beta_3 = -0.40$

Compare and comment on the relative hazard rates of Asian and African lions, when bred in captivity versus in the wild. (4)

[14]

- Q. 10)** A specialist insurer sells only term assurance policies through a number of banks. It has collected claims data from last ten years of operations and this is provided below. The insurer wishes to use the mortality rates based on its own experience for valuation purposes and also for pricing any new products in the future.

Age group [Age last birthday]	Initial exposed to risk	Number of deaths	Crude mortality rates
20-24	74	1	1.3514%
25-29	5,982	12	0.2006%
30-34	27,839	65	0.2335%
35-39	35,487	124	0.3494%
40-44	40,859	156	0.3818%
45-49	39,850	220	0.5521%
50-54	34,859	189	0.5422%
55-59	29,349	210	0.7155%

You have been asked to determine a set of graduated rates from the crude estimates provided above for use by the insurance company.

- a)** Study the data carefully, and state clearly whether the graphical method of graduation is appropriate in this case.

A graphical graduation has been carried out and you are provided with the following graduated rates:

Age group [Age last birthday]	Graduated mortality rates
20-24	0.1515%
25-29	0.2089%
30-34	0.2731%
35-39	0.3442%
40-44	0.4223%
45-49	0.5075%
50-54	0.6000%
55-59	0.7000%

The Appointed Actuary is concerned that individual bias may have led to the rates being either under- or over-graduated.

(6)

- b)** Clearly state the null hypothesis and carry out the chi-square in conjunction with another appropriate test to determine whether or not the crude rates are under- or over-graduated.

(10)

- c)** Comment on your conclusions from part (ii) and the appropriateness of using the chi-square test in this case.

(3)

[19]
